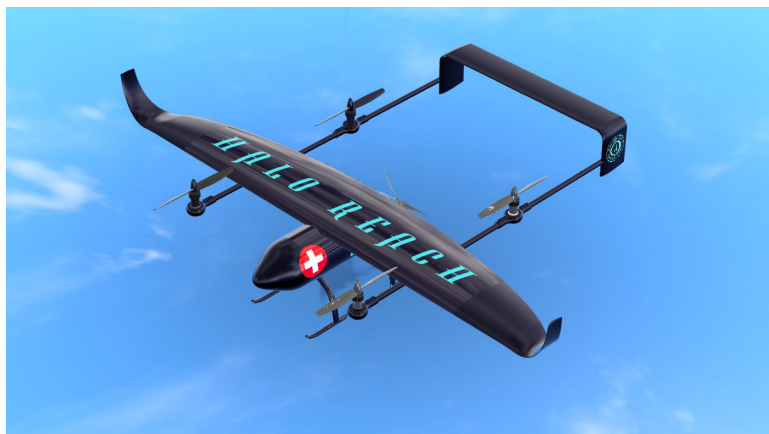


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## **HALO REACH: *Optimising Weight and Balance through Comprehensive Powerplant Sizing for VTOL Fixed-wing UAV***

UAS for Disaster Relief - AE3 2024

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### **Abstract**

This report outlines the selection and optimisation of the powerplant for the HALO REACH Project, designed for disaster relief missions. The UAV's primary missions include long-endurance flights with a 20 kg payload loitering for 10 hours and supply delivery with a 50 kg payload loitering for 20 minutes, over a range of 185 km at 500 meters altitude.

Various UAV configurations were assessed, focusing on fixed-wing VTOL designs for their efficiency. The powerplant options considered hybrid systems with Jet-A fuel offer higher energy density and range but add complexity.

The motor selection process identified the Herlea H140 as the optimal choice. Battery selection targeted high-energy-density technologies, with engagement from companies like AMPRIUS. The HIRTH 4202HF engine was chosen for its optimal balance of power output and weight, enhancing the UAV's overall efficiency.

Weight and balance optimisation ensured stability across mission profiles. A complex hybrid system was proposed to dynamically adjust the power source for enhanced efficiency.

Future recommendations include further research into advanced energy storage and systematic project management to optimise the UAV's performance and sustainability.

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## Nomenclature

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| $\alpha$                 | Angle of attack (degrees)                        |
| $\eta$                   | Merit Index (W/g)                                |
| $\kappa$                 | Correction factor in rotor theory                |
| $\rho$                   | Air density (kg/m <sup>3</sup> )                 |
| $A$                      | Total area of the multirotor (m <sup>2</sup> )   |
| $C$                      | Charge/discharge rate (C-rate)                   |
| $C_{D,v}$                | Vertical drag coefficient                        |
| $CG$                     | Center of Gravity                                |
| $D_v$                    | Vertical drag (N)                                |
| $DOD$                    | Depth of discharge )                             |
| $EW$                     | Empty Weight (kg)                                |
| $FM$                     | Rotor figure of merit                            |
| $GTOW$                   | Gross Takeoff Weight (kg)                        |
| $m_{\text{fuel}}$        | Mass of fuel (kg)                                |
| $MTOW$                   | Maximum Takeoff Weight (kg)                      |
| $MZFW$                   | Maximum Zero Fuel Weight (kg)                    |
| $N_t$                    | Number of fuel tanks                             |
| $N_{\text{en}}$          | Number of engines                                |
| $P_{\text{hover}}$       | Power required to hover (W)                      |
| $P_R$                    | Power required (W)                               |
| $R_{\text{int}}$         | Internal resistance (m $\Omega$ )                |
| $S_p$                    | Projected area of the aircraft (m <sup>2</sup> ) |
| $V_i$                    | Integral fuel tank volume (gallons)              |
| $V_t$                    | Fuel tank volume (gallons)                       |
| $V_{RoC}$                | Rate of climb (m/s)                              |
| $W_{\text{fuel system}}$ | Weight of the fuel system (kg)                   |
| $Wh/kg$                  | Watt-hours per kilogram (energy density)         |
| $x_{CG}$                 | CG position in x-direction (m)                   |
| $y_{CG}$                 | CG position in y-direction (m)                   |
| $z_{CG}$                 | CG position in z-direction (m)                   |

# 1 Introduction

## 1.1 Aims and Mission

The primary aim of HALO REACH is to support disaster relief air operations. The goal is to design a UAV configuration that can achieve the following mission profiles:

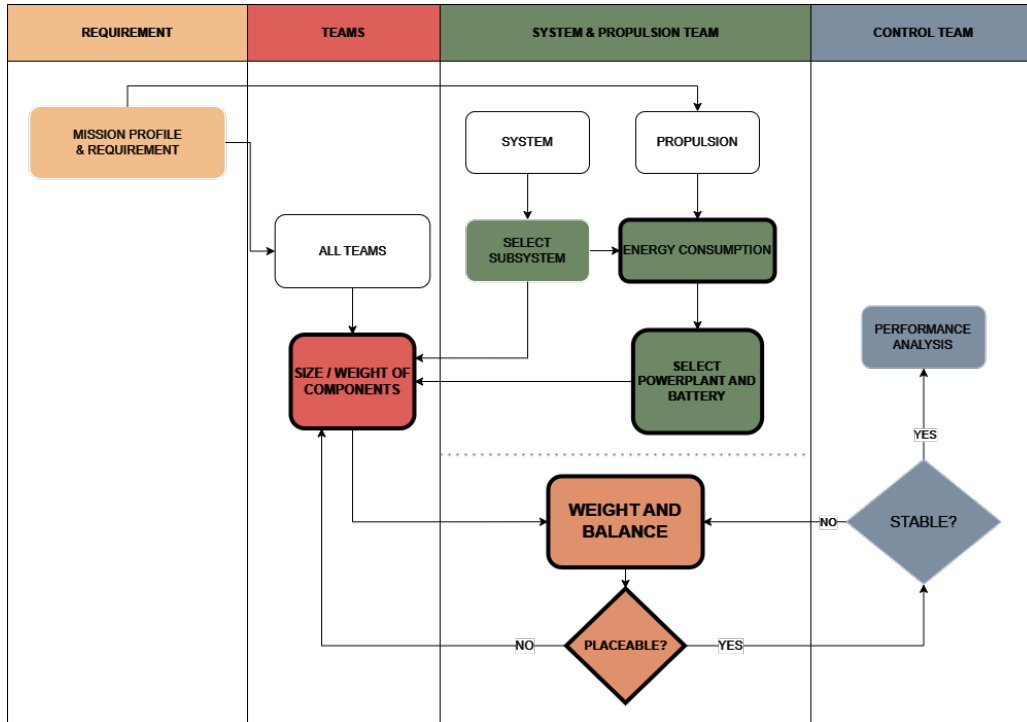
- **Long Endurance:** Carry a 20 kg payload with the capability to loiter for at least 10 hours.
- **Supply Delivery:** Carry a 50 kg payload with the capability to loiter for 20 minutes.

For both missions, the UAV must cover a range of 185 km and operate at a design altitude of 500 meters.

## 1.2 Team Responsibilities and Focus Areas

The internal systems and propulsion team, as detailed in the Project Coordinator's report [1] [2], is responsible for the selection and integration of propulsion systems, as well as overall system design. Within the systems and propulsion team, each sub-team has specific focus areas, primarily on propulsion and system selection. This report focuses on optimising the drone's weight and balance from a propulsion perspective, starting with the drone configuration through to powerplant sizing.

## 1.3 Overall Process



**Figure 3.** Flowchart showing team allocation and its impact on workflow

The overall process of weight and balance in HALO is illustrated in Figure 3, and this report will focus on the elements in bold within the flowchart. The aim is to ensure collaboration with the stability and control team, especially with Ruairidh's report [3], which combines all weight and positioning data from all teams to ensure the stability in collaboration with the control team.

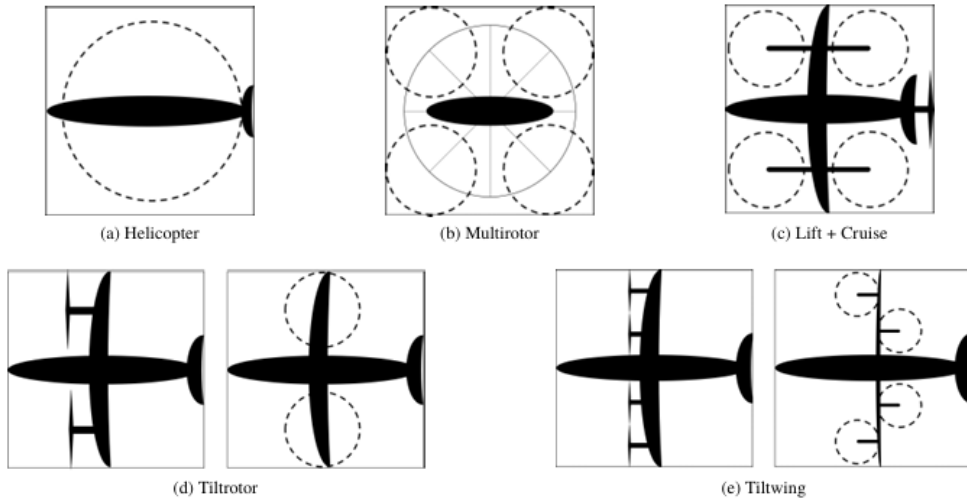


## 2 UAV Configuration

To address the mission profiles of disaster relief, selecting the optimal UAV configuration is one of the most significant steps in the process. For all missions, the drone requires the capability of VTOL and specifically be able to carry payloads of 20 kg for long endurance and 50 kg for supply delivery. After considering the most constrained case from the propulsion team’s perspective, the 10-hour endurance mission can be determined as the primary focus.

### 2.1 Main Configurations

The initial configurations that were capable of eVTOL considered included single-rotor (helicopters), multi-rotors, fixed-wing VTOL, tilted wings, and tilted rotor designs. Our primary concerns were the energy consumption and endurance of the UAV to meet the 10-hour requirement. Ensuring that the UAV could sustain long-duration missions while maintaining energy efficiency was crucial.

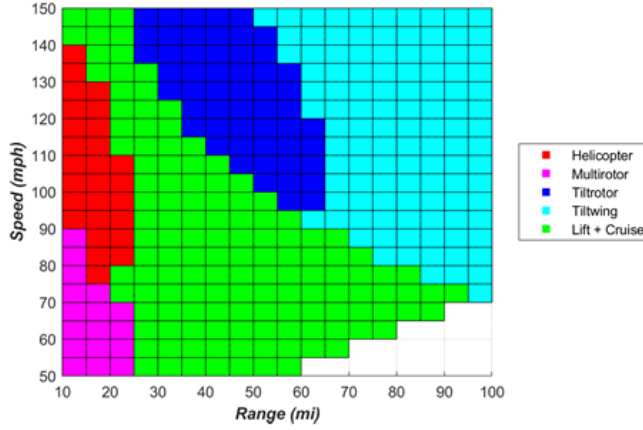


**Figure 4.** General conceptual layouts for candidate configurations [4]

Using a weights-based optimisation scheme, five primary eVTOL configurations were designed and evaluated for various range/speed mission profiles. The optimal configuration for each mission was determined based on the lowest feasible gross weight.

### 2.2 Performance Charts

Optimal configurations vary with cruise range and speed, shown in Figure 5. Winged configurations perform poorly at low ranges due to the added weight of wings, but excel at higher ranges and speeds. Non-winged configurations like multirotors and helicopters are limited in range and speed due to their lower efficiency in cruise. This suggests that wingless configurations are unlikely to achieve endurance of up to 10 hours due to their continuous rotor operation to generate lift. This results in high energy consumption, as the rotors must constantly work to keep the UAV aloft, leading to a rapid reduction of the battery or fuel supply. In contrast, a design with wings is more feasible for achieving long endurance. Wings generate lift through aerodynamic forces as the UAV moves forward, which is much more energy-efficient. This allows for longer flight times without the need for excess continuous rotor power. Consequently, designs with wings are better suited for missions requiring extended endurance and longer range.



**Figure 5.** Optimal configuration for different range/speed missions [4]

All the pros and cons of each configuration are summarised in Table 2. Based on the mission profile stated in section 1, the drone operates to cover a range of 185 km. The Tiltwing and Lift + Cruise (fixed-wing) configurations are the main candidates for selection. However, the fixed-wing configuration is more likely to be chosen as the pusher propeller can be modified to integrate an internal combustion engine, improving cruise efficiency. Moreover, the structural design of the Tiltwing is very complicated, posing challenges not only for the structural team but also for the control team during the transition phase. Therefore, the design chosen is the **VTOL fixed-wing configuration**.

**Table 2.** Pros and Cons of Different eVTOL Configurations [4][5]

| Configuration | Pros                                                                                                                                                                                 | Cons                                                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multirotor    | <ul style="list-style-type: none"> <li>- Optimal for short ranges and slow speeds.</li> <li>- Simple design and controllability.</li> </ul>                                          | <ul style="list-style-type: none"> <li>- Limited by propeller efficiency.</li> <li>- Inefficient for longer ranges and higher speeds.</li> </ul>                                                       |
| Helicopter    | <ul style="list-style-type: none"> <li>- The best performance in hover efficiency.</li> </ul>                                                                                        | <ul style="list-style-type: none"> <li>- Complexity and heavy compared to multirotor</li> </ul>                                                                                                        |
| Lift + Cruise | <ul style="list-style-type: none"> <li>- Most broadly optimal configuration.</li> <li>- Combines lift rotors and a pusher propeller for high cruise and hover efficiency.</li> </ul> | <ul style="list-style-type: none"> <li>- Outperformed by multirotors and helicopters at low speeds and ranges.</li> <li>- Outperformed by tiltwing and tiltrotor at high speeds and ranges.</li> </ul> |
| Tiltwing      | <ul style="list-style-type: none"> <li>- Best for long ranges and high speeds.</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>- Complicated mechanism.</li> </ul>                                                                                                                             |
| Tiltrotor     | <ul style="list-style-type: none"> <li>- Slightly lighter compared to tiltwing due to smaller tilting mechanisms.</li> </ul>                                                         | <ul style="list-style-type: none"> <li>- Less efficient in cruise compared to tiltwing.</li> <li>- Tilting mechanism needed</li> </ul>                                                                 |

### 3 Energy Consumption

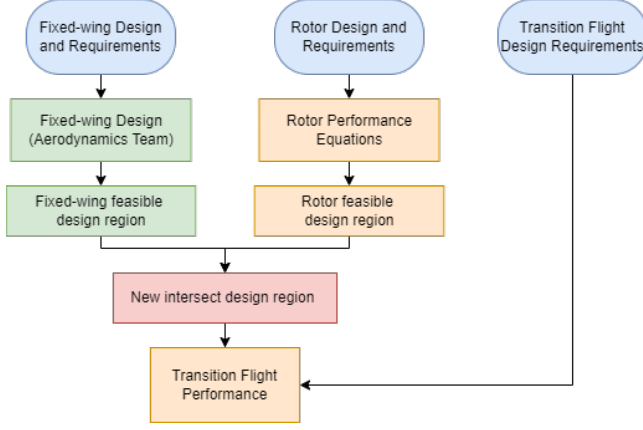
#### 3.1 Power System Considerations

After being designed to be a VTOL fixed-wing aircraft, the design now has two main considerations:

- Fully Electric: All systems operate with electrical power.
- Hybrid Electric: Jet-A fuel is involved in the power system.

The ideal case is to have a fully electric drone or eVTOL fixed-wing due to sustainability and noise constraints. Fully electric systems are not only sustainable but also fully rechargeable, reducing the dependency on fossil fuels. However, the energy density of today's battery technology is significantly lower compared to Jet-A fuel, with 0.5 MJ/kg for batteries compared to 43.3 MJ/kg for the fuel [6]. This disparity suggests that fully electric UAVs currently face limitations in terms of range and endurance.

On the other hand, a hybrid electric system, which involves the use of jet-A fuel, offers more power and a longer range due to the higher energy density of the fuel. However, this approach requires the integration of an internal combustion engine, which introduces complexities such as increased maintenance, higher noise levels, and less sustainability.



**Figure 6.** VTOL Design Process

Therefore, a feasibility analysis has been conducted based on the mission profile to determine the most practical and efficient power system for the UAV. The Figure 6 shows the process of calculating and designing the fixed-wing and rotor parts for the VTOL phase to the transition phase. For both mission profiles mentioned in section 1, the UAV requires both rotor mode and fixed-wing mode. While the fixed-wing design has been made by the aerodynamic team [7], this report aims to explain the approach to calculate the power requirements for the rotor mode (VTOL Phase).

### 3.2 VTOL Phase

In the VTOL phase, both the total weight  $W_{TO}$  and the vertical drag  $D_v$  of the aircraft must be countered by the rotor-generated thrust  $T_v$ . This required thrust  $T_v$  is calculated as:

$$T_v = 1.15 \times (W_{TO} \pm D_v) \quad (1)$$

where factor 1.15 accounts for the additional thrust needed to overcome wind effects [8],  $-D_v$  occurs when landing.

The vertical drag  $D_v$  is given by:

$$D_v = \frac{1}{2} \cdot C_{D,v} \cdot \rho \cdot v_v^2 \cdot S_p \quad (2)$$

where  $\rho$  is the air density,  $C_{D,v}$  is the vertical drag coefficient,  $v_v$  is the vertical velocity during take-off, and  $S_p$  is the projected area of the aircraft. For VTOL, the drag coefficient  $C_{D,v}$  is  $2 \cdot \sin^3(\alpha)$ . Since  $\alpha$  is  $90^\circ$  during VTOL,  $C_{D,v}$  simplifies to 2 [9].

Using the momentum theory, the power  $P_R$  required to generate the thrust  $T_v$  is calculated by: [10]

$$P_{hover} = T_v \sqrt{\frac{T_v}{2\rho A}} / (FM) \quad (3)$$

where A is the total area of the multirotor and  $FM$  is the rotor figure merit.

$$FM = \frac{\text{Ideal power required to hover}}{\text{Actual power required to hover}} < 1 \quad (4)$$

Using the modified form of the momentum theory with the non-ideal approximation for the power, the figure of merit can be written as [11]

$$FM = \frac{\text{Ideal power}}{\text{Induced power} + \text{profile power}} = \frac{P_{ideal}}{\kappa P_{ideal} + P_0} = \frac{2}{3\kappa} \quad (5)$$

where  $\kappa$  is the correction factor used in rotor theory, estimated to be 1.15 for the preliminary design. The rate of climb,  $V_{RoC}$ , was designed to be 1.0 m/s for takeoff and -0.5 m/s for landing. These values were considered based on the requirements for disaster relief missions, where lifting humans is a potential scenario. Having too high a climb rate could affect the stability of the flight. The calculations were initially made to determine the power needed for the motor using the equation:

$$P_R = P_{hover} + \frac{3}{2} mg V_{RoC} \quad (6)$$

These values were later verified and refined by the aerodynamics team [12] to derive the designed propeller parameters and obtain the exact numbers, including drag. The results of both calculations are shown in Table 3. The plot can be found in Appendix A

**Table 3.** Comparison of Drag and Power Required

|                     | Drag (N) | Max Power Required (kW) |
|---------------------|----------|-------------------------|
| Initial Calculation | 7.03     | 44.84                   |
| Refined Model       | 3.91     | 40.8                    |

### 3.3 Other Phases

After stably hovering at the altitude of 10m, the UAV then undergoes the transition phase, where the power used by the rotors to produce lift decreases as horizontal speed increases, driven by the pusher engine. In this phase, the drone changes from relying mainly on the vertical lift by the rotors to the aerodynamic lift produced by the fixed wings as the aircraft moves forward. During the initial stage of the transition, the rotors generate lift to remain hover. However, as the pusher engine accelerates the UAV forward, the wings begin to produce aerodynamic lift. Consequently, the lift needed from the rotors decreases gradually until the fixed wings generate sufficient lift to maintain the UAV's altitude without the need for rotors' lift.

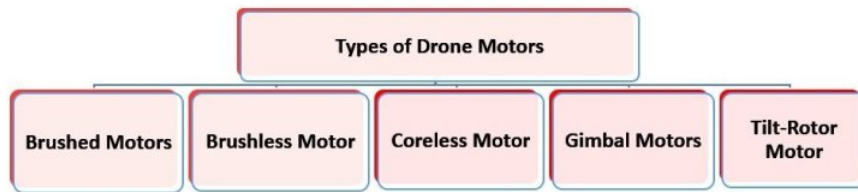
The UAV's climb, cruise, and loiter phases are now powered by the engine, which is optimised for efficient flight in these modes. The calculation of these phases will be mentioned in George Halim's report [13]. The plot of the overall power required is shown in Figure 13 in Appendix A

## 4 Powerplant Initial Selection

The powerplant selection has been conducted based on products available on the online market. Most eVTOL drone products, especially motors, have been manufactured and customised to optimise their performance. This report outlines the selection process based on existing options and identifies the product that is most optimised and well-suited for the HALO REACH mission.

### 4.1 Motor Selection

To satisfy the requirement for the power output in the VTOL phase, a motor that can produce a power output of 11.21 kW is required. The motor specification should have enough power to lift approximately 40 kg per motor. One of our first choices comes from the brushless motor type used in heavy-lift drones [14], particularly those designed for large payloads. These motors commonly leverage electric brushless DC technology. Brushless motors are preferred over brushed motors due to their higher efficiency. In this motor, the transfer of current from the commutator to the rotor occurs through non-contact methods, reducing energy losses associated with friction.

**Figure 7.** Common types of motors [14]

Initially, a few brushless motor brands were researched, such as T-motor, Mechtex, and KDE Direct. For the GTOW of 160 kg, the U-Type motors from the T-Motor company were considered due to their specific design for heavy-lift drones. A few models of U motors are shown in Table Table 4. To optimise the selection, the performance of the motor was evaluated based on the merit index.

$$\eta = \frac{\text{Max Power}}{\text{MASS}} \quad (7)$$

**Table 4.** Comparison of different T-motor U-type models

| Model           | Weight (kg) | Price (\$) | Max Power (kW) | $R_{int}$ (m $\Omega$ ) | $\eta$ (W/g) |
|-----------------|-------------|------------|----------------|-------------------------|--------------|
| U15II KV80 [15] | 1.74        | 689.90     | 11.44          | 17.00                   | 6.57         |
| U15L 43KV [16]  | 3.60        | 1099.00    | 15.51          | 20.00                   | 4.31         |
| U15XL 38KV [17] | 4.41        | 1699.00    | 18.31          | 17.00                   | 4.15         |

The U15II KV80 seemed to be the best candidate among the three motor models due to its merit index and being the cheapest. However, considering the hover phase, the motor needs to perform at its maximum power continuously for more than 10 seconds (VTOL to 10 m with rate of climb 1 m/s) [12]. This would lead to a significant loss in efficiency due to factors such as heat loss [15]. It was concluded that the motor was too close to its limits, as the safety factor is aimed to be about 1.5 [18]. Therefore, higher-power motors were considered, such as the U15L 43KV and U15XL 38KV. Due to a better merit index, the **U15L 43 KV** was initially chosen.

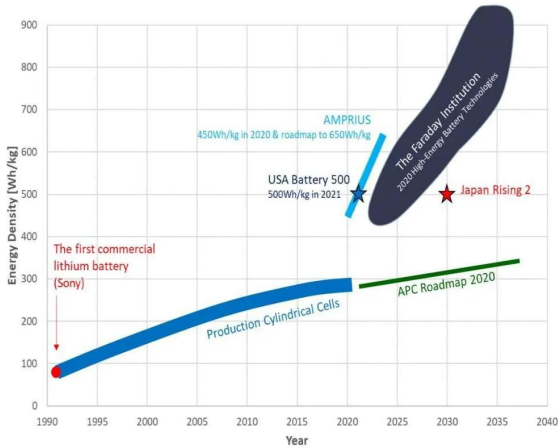
## 4.2 Battery Selection

The selection of batteries for a VTOL UAV involves several key factors beyond just weight. These factors include safety and certification, cost, and charging infrastructure.

### 4.2.1 Safety and Certification

Our project objective is disaster relief, and prioritising safety of battery safety is essential, especially for Unmanned Aerial Vehicles (UAVs). Compliance with aviation standards such as Part 23 of the FAA, EASA, and JCAB regulations is important to ensure the safety of the UAVs. Part 23 regulations address battery safety through measures like containment (14 CFR § 23.1353(b)) to prevent overheating and explosions, and mitigation of hazardous gas accumulation (14 CFR § 23.1353(d)) in case of malfunctions. The selected batteries must meet the criteria to be considered airworthy.[2]

### 4.2.2 Weight and Energy Density



Weight is a crucial factor in battery selection, directly impacting the UAV's overall performance and endurance. Batteries with higher energy density (measured in Wh/kg) are preferred as they provide more energy for the same weight. Currently, lithium-ion batteries are commonly used due to their favorable energy density, but future trends indicate the potential of silicon anode and solid-state batteries and other advanced technologies to offer even higher energy densities as seen in Figure 8.

**Figure 8.** Cell Energy Density Roadmaps [19]

REACH, our UAV, was designed based on technology from 2024 onwards. As the development of battery technology faces difficulties due to material limitations, most companies have not reached their target plans and are still in the phase of developing and testing before commercialisation. Lithium-ion technology still dominates the battery market. However, AMPRIUS technology launched the product AMPRIUS SiMAXX with a battery density of up to 500 Wh/kg in 2021 [19] and successfully made it commercial in 2023. Using the Silicon Nanowire Anode, it enhances capacity by 10 times compared

to graphite [20], typically used as the material for the anode with lithium. They predict the battery density to reach 650 Wh/kg by 2025.

Although the next step in battery technology beyond silicon anodes, such as solid-state lithium metal, is being developed by companies like Electric Power Systems [21], interface stability and material complexity need to be addressed before commercialisation. Therefore, the emerging advanced performance in silicon anode technology makes the battery **AMPRIUS SIMAXX** the top choice and was chosen.

#### 4.2.3 Charging Infrastructure

Charging infrastructure is another important aspect of battery selection. Efficient charging systems must be in place to support the efficiency of times between missions. This includes the availability of charging stations and the ability to handle the electrical load required for rapid battery recharging. The infrastructure should also consider safety measures to prevent overcharging and overheating, ensuring the longevity and reliability of the batteries.

The AMPRIUS SIMAXX has 3 designs: High power, High energy, and Balanced. Their performance can be found in Table 9 in the Appendix B.1 Based on power requirements and discharge rates, the selected batteries must provide sufficient energy while maintaining a balance between power and performance. Considering the mission profile, which includes high power needs for supply delivery missions and high energy needs for long endurance missions, the Balanced Design battery was chosen. With an energy density of 450 Wh/kg, a charging rate of 4C (80% in 15 minutes), and a discharge rate of 4C continuous / 6C pulse, it offers an optimised balance for both high-power and long-endurance missions. Additionally, its cycle life of more than 300 cycles at 80% DOD ensures durability and reliability. Overall, the battery selection process involves a comprehensive evaluation of multiple factors to ensure the optimal performance and safety of the VTOL UAV.

#### 4.2.4 Cost

The cost of batteries is a significant consideration. High-performance batteries with superior energy density and longevity tend to be more expensive. The cost must be balanced against the benefits they provide in terms of endurance and performance. Additionally, the total cost of ownership, including maintenance and replacement costs, should be evaluated which is stated in the project's coordination reports [1]

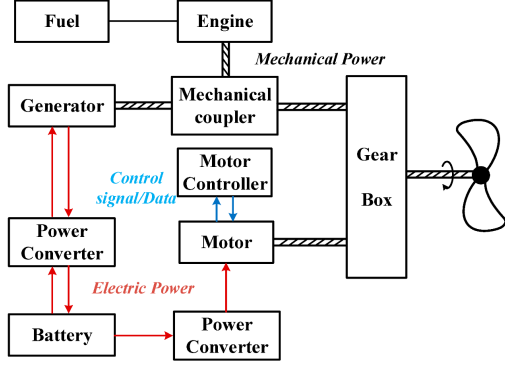
### 4.3 Engine Selection

In George's report [13], it was shown that a fully electric UAV with a battery density of 450 Wh/kg and the aerodynamic parameters such as lift-to-drag ratio [7], was not able to complete the long-endurance mission. Therefore, as mentioned in section 2, the fixed-wing lift and cruise configuration has the advantage of integrating an engine. Thus, the engine selection becomes a significant consideration. The main considerations are weight, compatibility of internal combustion engine (ICE) and Jet A fuel, and optimal power generation. Internal combustion engines designed for Jet A-1 fuel are rare because Jet A-1 suits turbine engines used in aircraft, which operate under different conditions. Jet A-1's properties, like its high flash point and specific energy content, are ideal for the high temperatures and pressures in jet engines but not suitable for standard piston engines that run on gasoline or diesel [22]. Research has been conducted, and George's report mentions details of engine types including two-stroke and Wankel engines. The engine initially chosen is the HIRTH 4202HF. The key parameters are shown in Table 10 in Appendix B.2.1.

The engine has the option to integrate a generator, adding 0.8 kg to generate 1 kW. The performance analysis for transition, cruise, and loiter states was calculated in the aerodynamics team report [23] based on this engine's performance.

The engine, weighing 11.5 kg, significantly contributes to shifting the CG backwards, making the UAV more unstable. However, the integrated generator, which can further reduce the mass of the battery, was a good trade-off. Therefore, the engine **HIRTH 4202HF** was initially chosen and used for the pusher propeller.

#### 4.4 Electrical and Fuel System



**Figure 9.** Complex Hybrid System

When the internal combustion engine (ICL) is operating in our UAV, the fuel system becomes crucial. A hybrid system has been introduced, and there are several approaches to integrate it, such as parallel, series, series-parallel hybrid, and complex hybrid [24]. The complex hybrid was chosen due to its advantages. Components such as the converter, gearbox, and generator have already been selected by the system team and the aerodynamic team. The fuel system that corresponds with the Hirth engine needs to be chosen to achieve the most optimal weight.

The weight of the fuel system can be calculated using the following equation [25]:

$$W_{\text{fuel system}} = 2.49V_t^{0.726} \left( \frac{1}{1 + V_i/V_t} \right)^{0.363} N_t^{0.242} N_{\text{en}}^{0.157} \quad (8)$$

Where:  $V_t$  is the fuel tank volume in gallons,  $V_i$  is the integral fuel tank volume in gallons,  $N_t$  is the number of fuel tanks, and  $N_{\text{en}}$  is the number of engines.

The weight is calculated to be 6.12 kg for the entire fuel system, including the fuel tank, fuel pump, filter, and sensor. However, the equation is based on the transport fixed-wing aircraft. This number is used as a maximum constraint weight for the overall fuel system.

Looking more closely at the components, REACH requires a maximum fuel capacity of 36.5 kg, which is around 43.5 L (Jet-A density of 840 kg/m<sup>3</sup> [26]). The fuel tank for the UAV can use **bladder tanks from ATL** company [27], which are commonly used in military-grade applications. These tanks are very lightweight and can operate Jet A in the temperature range of  $-50^\circ\text{C}$  to  $95^\circ\text{C}$ . Therefore, the weight of the fuel tank is significantly decreased.

For the fuel pump, the size of the fuel tank significantly affects the size of the fuel pump due to the different pressure requirements. The typical weight of commercial fuel pumps for UAVs also depends on the size of the UAV. REACH is considered to be in the large category with 43.5 L of Jet-A. The fuel pump is approximately 0.7 to 2.0% of the MTOW [24]. The fuel pump **Walbro FRB-13** has been selected as it is specifically designed for UAVs and has a high flow rate. The weight is 0.83 kg [28], which is lower than the estimated value.

Apart from this, components such as the fuel filter, fuel lines, and sensors are chosen by the Hirth company to work best with their engine.

Therefore, for the UAV, and considering new technologies, the weight of the fuel system has been constrained to 5.0 kg.

## 5 Weight and Balance

All previous sections were designed to initially size REACH for weight and balance. The power-plant selection significantly impacted the weight and energy resources. After obtaining all component weights and sizes from all teams, the weight and balance optimization was conducted with the aim of achieving a 160 kg MTOW, as seen in Figure 3. The main target with the control team is to ensure stability by having the X-CG in front of the neutral point. The interaction of designing the tailwing will run parallel to this process and will be explained in more detail in Ruairidh’s report [3].

Moreover, different scenarios need to be considered, such as supply delivery and long-endurance cases with different payloads. Without payload, the CG shift could make the flight unstable. Therefore, the significance of weight and balance strategies focus on positioning the majority of the weight as close as possible to the CG. Additionally, for fuel, whose weight changes during flight, the same strategies need to be applied to avoid overshifting of the CG.

At the early stage, with all design and sizing, the UAV faced a significant overweight problem, so some powerplant components needed to be re-selected.

### 5.1 Process of Resizing

The goal of every team was to get lighter components with optimised performance while still maintaining safety. The resizing process of the propulsion team needed to consider the effect on performance analysis, which might further affect other design processes. Therefore, the new components should prioritise similar or better performance and then be lighter in weight.

#### 5.1.1 New Motor Selection

After the iterative process and including more drone components, the weight of 3.60 kg for each motor was significant and massively increased the thickness of the boom connecting them, thereby adding more total weight. The motor weight needed to be reduced, and based on similar performance, the research was conducted and the Herlea H140 motor model was selected.

**Table 5.** Specification of Herlea H140 KV80 and KV95 [29]

| Motor Type | Weight (kg) | Price (\$) | Max Power (kW) | R <sub>int</sub> (mΩ) | η (W/g) |
|------------|-------------|------------|----------------|-----------------------|---------|
| KV80       | 2.57        | 919.00     | 12.07          | 16.00                 | 4.70    |
| KV95       |             |            | 13.20          | 21.00                 | 5.14    |

The motor was finally selected to be the **Herlea H140 KV95** due to its high merit index and a motor safety factor of 1.30 compared to the power needed for the refined model Table 3.

#### 5.1.2 Engine Consideration

Although the performance and flight analysis has been done with the HIRTH engine, most components could not fit in the weight limit of 160 kg. After considering the compatibility of the Wankel engine type mentioned in George’s report [13], research was conducted focusing on Wankel engines that can use Jet A fuel and produce the same power of 11 kW with lighter weight. The UAVHE RW79 engine was found to be suitable. The new engine’s key parameters were compared as shown in Table 6.

**Table 6.** Comparison of Key Parameters of Each Engine

| Engine       | Type     | Max Power         | Weight (kg) | Dimensions (mm) |
|--------------|----------|-------------------|-------------|-----------------|
| HIRTH 4202HF | 2-Stroke | 11 kW @ 6500 rpm  | 11.50       | 215 x 334 x 365 |
| UAVHE RW79   | Wankel   | 16 kW @ 13500 rpm | 4.71        | 177 x 197 x 162 |

Looking only at performance and the weight of the engine, the Wankel engine seems to be well-suited and the overweight problem would be immediately solved. The RW79 engine was clarified as an APU,



generating only electrical power. The contact has been made with the company, and the propeller version can be adapted, as shown in Figure 14b in Appendix B.2.2, with our designed RPM. So, the addition of mechanisms such as a motor or control system that would increase complexity and weight is not necessary added. This change would not affect any previous performance and would even result in improvements.

Nevertheless, changing the engine at a later stage of the project needs to be decided very carefully with all teams. Changing the weight of the engine would massively shift the position of the drone components, which were previously stabilised. Moreover, the refined model with the new optimised weight from the structure team suggests a massive reduction in the weight of the aircraft frame, including the wing, fuselage, empennage, and landing gear. This results in the available weight for the same engine to be fitted in.

## 5.2 Weight Estimation

Most of the component weights have been determined by different teams. The main weight optimisation comes from the structure and add-on teams, where the SP team obtains the weight from research and selection [30][31]. All the powerplant components have been selected and summarised in Table 7

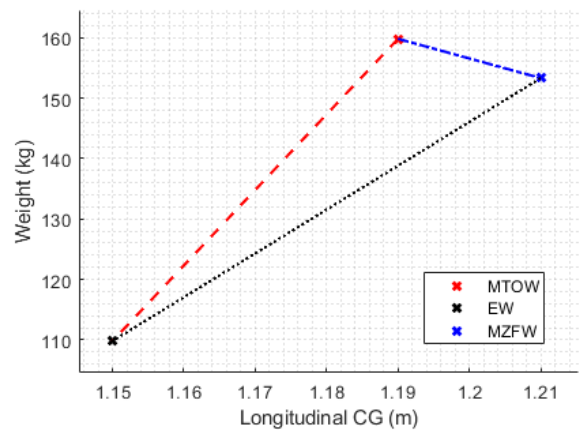
**Table 7.** Final powerplant selection

| Type    | Choice                    | Important specifications                                                |
|---------|---------------------------|-------------------------------------------------------------------------|
| Engine  | Hirth 4202HF              | - Max power output 11 kW,<br>- 11.5 kg mass with 0.8 kg, 1 kW generator |
| Motor   | T-Motor H140 KV95         | Max 13.2 kW output, meeting requirements<br>- 2.57 kg mass              |
| Battery | Amprius SIMAXX , Balanced | - Energy density 450 Wh/kg                                              |

All components in the REACH UAV are shown in Table 8. The supply-delivery mission is the constraint case for the weight of the UAV. In the long-endurance case, the payload is 20 kg, and the 30 kg gap between the different payloads needs to be managed, as the UAV is designed with optimised parameters at 160 kg MTOW.

In George’s report [13], the supply-delivery case uses 6.5 kg of Jet A fuel. Therefore, for long endurance, there will be room for up to 36.5 kg of fuel, but only 18.15 kg is calculated to be needed to complete the 10-hour duration as specified in the mission profile.

All cases of all mission profiles were considered. The CG shift in the supply delivery case is particularly important due to the massive shift that occurs after dropping the 50 kg payload, leading to the most constrained scenario. The forward and rear limits of the CG are determined by the Empty Weight (EW) and Maximum Zero Fuel Weight (MZFW), respectively. The CG shift across all scenarios is illustrated in Figure 10. The shift of over 6 cm has been confirmed with the stability teams to ensure that the UAV remains stable, maintaining a static margin of less than 20%. This thorough analysis and confirmation process are essential for ensuring the UAV’s safe and efficient operation across different mission profiles.

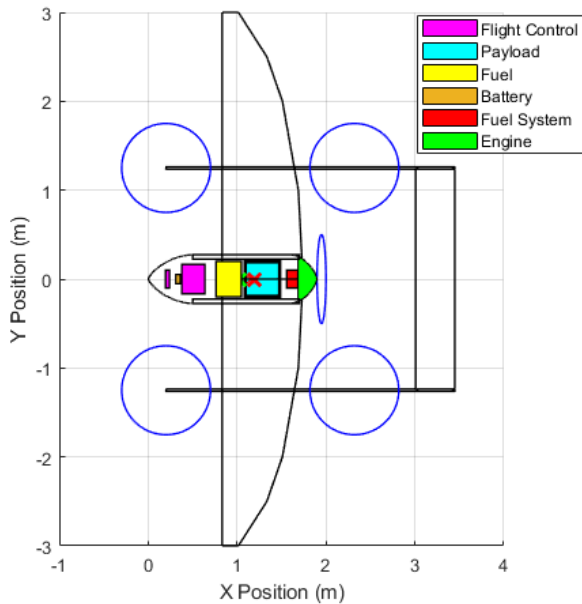


**Figure 10.** Longitudinal CG shift across cases

**Table 8.** Component weight with CG coordinates and assigned teams.

| Component      | Weight (kg)   | CG - x (m)  | CG - y (m)  | CG - z (m)  | Team                      |
|----------------|---------------|-------------|-------------|-------------|---------------------------|
| Wing           | 14.94         | 1.15        | 0.00        | 0.24        | Structure                 |
| Empennage      | 2.98          | 3.23        | 0.00        | 0.55        |                           |
| Back Propeller | 0.56          | 1.90        | 0.00        | 0.00        |                           |
| Fuselage       | 15.10         | 0.95        | 0.00        | 0.00        |                           |
| Right Boom     | 1.66          | 1.83        | 1.25        | 0.24        |                           |
| Left Boom      | 1.66          | 1.83        | -1.25       | 0.24        |                           |
| Landing Gear   | 3.50          | 1.10        | 0.00        | -0.40       | System<br>& Propulsion    |
| Battery        | 5.40          | 0.34        | 0.00        | 0.00        |                           |
| Jet A Fuel     | 6.50          | 0.90        | 0.00        | 0.00        |                           |
| Engine         | 11.50         | 1.79        | 0.00        | 0.00        |                           |
| Generator      | 0.80          | 1.79        | 0.00        | 0.00        |                           |
| Motor-Prop 1   | 3.13          | 0.20        | 1.25        | 0.24        |                           |
| Motor-Prop 2   | 3.13          | 0.20        | -1.25       | 0.24        |                           |
| Motor-Prop 3   | 3.13          | 2.32        | 1.25        | 0.24        |                           |
| Motor-Prop 4   | 3.13          | 2.32        | -1.25       | 0.24        |                           |
| SatComm        | 10.20         | 0.51        | 0.00        | 0.00        |                           |
| Flight Control | 7.38          | 0.22        | 0.00        | 0.00        |                           |
| Wiring System  | 3.10          | 1.50        | 0.00        | 0.00        | Add-On<br>Mission Profile |
| Fuel System    | 5.00          | 1.62        | 0.00        | 0.00        |                           |
| Add On System  | 7.00          | 1.29        | 0.00        | 0.00        |                           |
| Payload        | 50.00         | 1.29        | 0.00        | 0.00        | Add-On<br>Mission Profile |
| <b>Total</b>   | <b>159.80</b> | <b>1.19</b> | <b>0.00</b> | <b>0.05</b> |                           |

### 5.3 Structure Layout

**Figure 11.** Top View

From Figure 3, after all the components were chosen, their placement availability needed to be considered, with most components being in the fuselage. The dimensions of all components were modified as boxes with the maximum dimensions, to allow for some staging orientation in the fuselage. This process was done closely with the stability team when components needed to be adjusted for stability.

Initially, the fuel tank was considered to be placed in several locations, such as the wing, and to have 2 fuel tanks. One is the static fuel tank used to compensate for the 30 kg mission profile at the front to shift the CG forward. The other was at the back, actually being used for the engine. This idea was considered because stability increases as fuel is used. However, this method required complexity in the fuel system, and having 2 fuel tanks significantly increased the mass of the fuel system. Additionally, a heat insulator would need to be added for the rear fuel tank as it is close to the engine.

After considering these factors and automatically moving components to ensure stability, the optimal position was determined as shown in Figure 11. The side view can be found in Appendix C.1

## 6 Conclusion and Improvement

In summary, this report covers the iterative powerplant selection and sizing process to optimize weight and balance within the 160 kg limit. The initial selection was based on available options in the online market, which could be further refined with more time and research.

- For the motor, performance analysis should consider modifiable motors that can be customized to optimize performance for our mission profile, avoiding overpowering and minimising weight.
- For the battery, it would be beneficial to engage in detailed discussions with companies at the forefront of high energy density technology, such as AMPRIUS or the Faraday Institution. Considering that REACH drone manufacturing is planned for the future, better batteries can be incorporated, significantly improving performance.
- For engine, the RW79 engine offers a significant advantage in terms of weight and balance, dropping 6.8 kg. Furthermore, RW79 is in the process of developing an integrated fuel system that includes ignition control, fuel pump, cooling, etc. [32]. The total weight of 3.37 kg further decreases the initial number of fuel system components by 5 kg.

Even though flight efficiency and endurance increase as the mass is reduced from fuel burn, using fuel will be less of a priority in HALO. A trade-off analysis of fuel can be performed by reducing the amount of fuel and increasing the amount of battery. A smaller fuel system, due to lower fuel requirements, will also contribute to weight reduction. In the study, although the current energy density gap between battery and jet A resources seems high, emerging trends such as bio-fuel or hydrogen power, which has significantly higher energy density, need to be tracked. Therefore, not only weight needs to be considered but also other factors such as cost and complexity. Additionally, the trade-off of 30 kg for each mission profile could be considered to allow more swapping in battery capacity. With the complex hybrid system, the electrical power required to operate the pusher can be studied further. This study will become more significant as HALO, designed for disaster relief, must also consider environmental impact, improving sustainability, and reducing fuel usage.

In terms of weight and balance, a spreadsheet was created for everyone to update parameters, and a MATLAB function was developed to automatically update the CG calculation once the spreadsheet is updated. Despite these tools, miscommunication about weight updates still occurred, linking to a lack of concise stability predictions. To accomplish the project within five weeks, component weights should be frozen and confirmed for the control and stability team as soon as possible. An accessible workspace for all parameters, such as GitHub, should be considered to improve work efficiency and complete the project within the planned schedule. All of these factors should be taken into account to run the project in a more systematic and automated way.

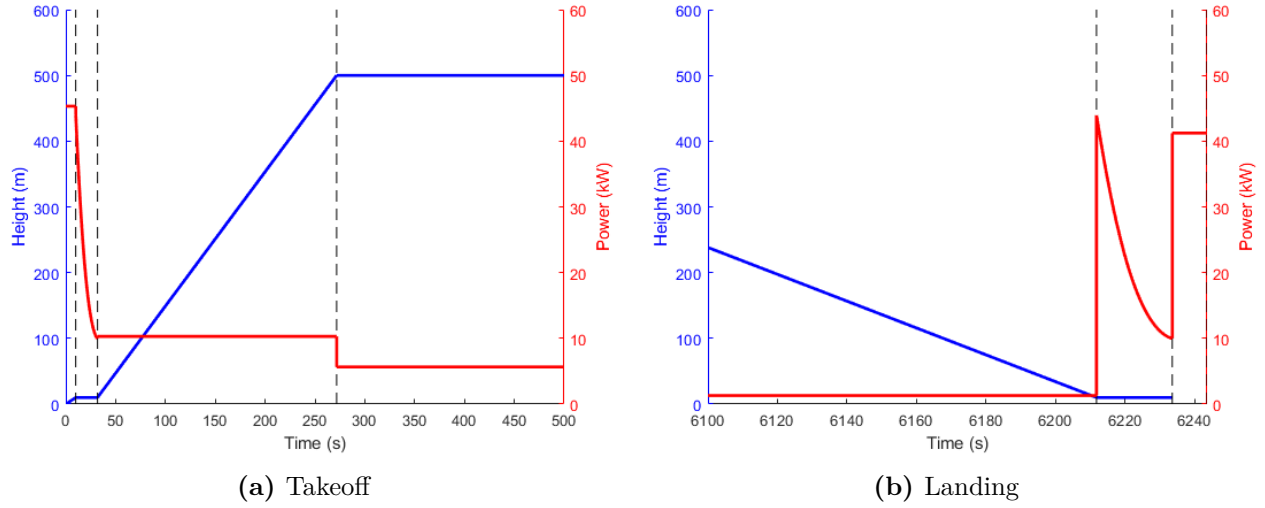
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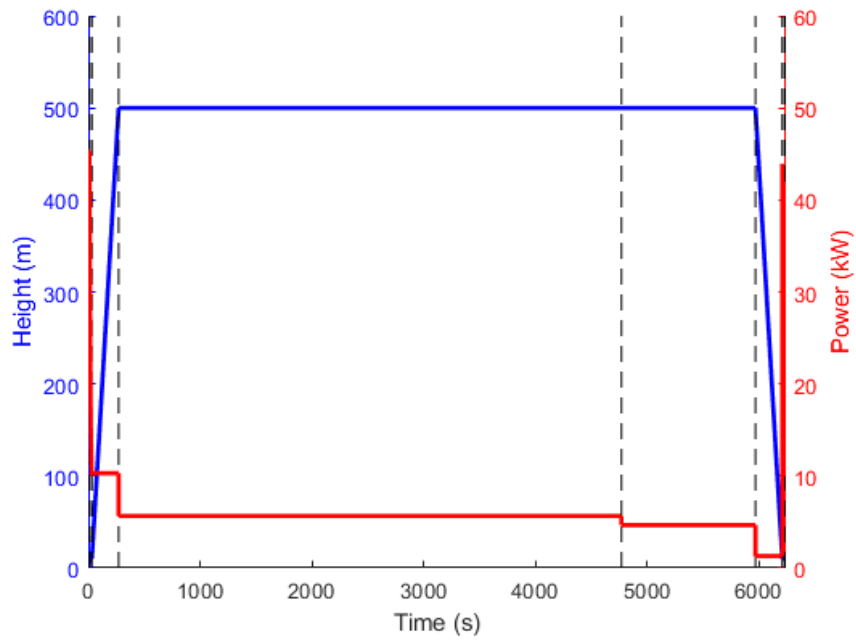
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# Appendix

## A Energy Calculation



**Figure 12.** Power requirements in VTOL phase



**Figure 13.** Power Requirements under supply-delivery mission

## B Powerplant Performance

### B.1 Battery

**Table 9.** Performance of Different SiMaxx Battery Designs [33]

| Design      | Energy Density<br>(Wh/kg) | Rate                           |                                   | 80% DOD<br>(Cycles) |
|-------------|---------------------------|--------------------------------|-----------------------------------|---------------------|
|             |                           | Charging<br>(max, 80% in mins) | Discharge<br>(Continuous / Pulse) |                     |
| High Power  | 400                       | 10C, 6 mins                    | 4C / 12C                          | 500                 |
| High Energy | 500                       | 2C, 30 mins                    | 1C / 3C                           | 200                 |
| Balanced    | 450                       | 4C, 15 mins                    | 4C / 6C                           | 300                 |

### B.2 Engine

#### B.2.1 HIRTH 4202HF

**Table 10.** Key Parameters of HIRTH 4202HF Engine [34]

| Type                 | Max Performance   | Weight (kg) | Dimensions (mm) | Fuel System    |
|----------------------|-------------------|-------------|-----------------|----------------|
| Two-Stroke (Opposed) | 11 kW at 6500 rpm | 11.5        | 215 x 334 x 365 | Fuel Injection |

#### B.2.2 UAVHE RW79



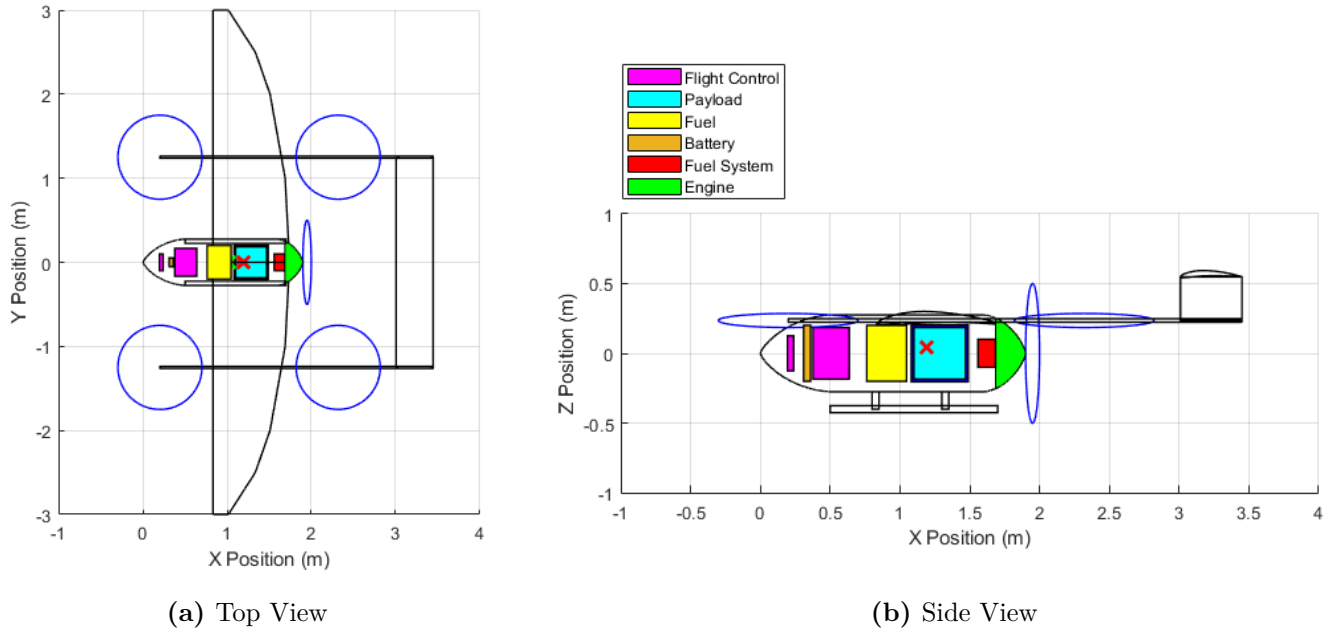
(a) UAVHE RW79

(b) UAVHE RW79 with Propeller

**Figure 14.** The UAVHE RW79 Variatiom

## C Weight and Balance

### C.1 Simplified View



**Figure 15.** Simplified view of the UAV structural Layout